

ALGORITHMS AND PROBLEM SOLVING USING C++

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Internship Project Report

OBJECTIVE

The objective of this project is to implement sorting , searching , and greedy algorithms using C++ and analyze their performance.

ALGORITHMS USED

- 1.) Bubble sort
- 2.) Binary search
- 3.) knapsack

PROGRAM CODE

```
#include <iostream>
using namespace std;

void bubbleSort(int arr[], int n) {
    for(int i=0;i<n-1;i++) {
        for(int j=0;j<n-i-1;j++) {
            if(arr[j] > arr[j+1]) {
                swap(arr[j], arr[j+1]);
            }
        }
    }
}

int main() {
    int arr[] = {5,2,9,1,7};
    int n = 5;

    bubbleSort(arr,n);

    for(int i=0;i<n;i++) {
        cout << arr[i] << " ";
    }
}
```

```
}  
  
    return 0;  
}
```

OUTPUT:

```
PS C:\Users\HP\OneDrive\Desktop\Algorithms_Project> cd C:\Users\HP\OneDrive\Desktop\Algorithms_Project\ ; if ($?) { g++ sorting.cpp -o sorting } ; if ($?  
?) { .\sorting }  
Sorted Array: 1 2 5 7 9  
PS C:\Users\HP\OneDrive\Desktop\Algorithms_Project>
```

TIME COMPLEXITY:

Bubble Sort:

Best Case: $O(n)$

Worst Case: $O(n^2)$

Binary Search:

Time Complexity: $O(\log n)$

Knapsack:

Time Complexity: $O(nW)$

CONCLUSION:

This project helped in understanding sorting, searching, and greedy algorithms using C++. The implementation improved programming and problem-solving skills.